



# Saliva & Swallowing — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

## HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

**4 Easy · 10 Medium · 6 Hard** · suggested time ≈ 25 minutes.

## Q Questions

**1. A healthy volunteer has been sitting quietly and has not eaten for several hours. Which gland provides most of his unstimulated saliva?**

EASY

- A. Parotid gland
- B. Parotid and sublingual glands in equal parts
- C. Submandibular gland
- D. Sublingual gland
- E. Minor salivary glands

**2. A 7-year-old chews a piece of plain bread for a minute. Which salivary component has begun digesting its starch?**

EASY

- A. Mucins
- B. Lactoferrin
- C.  $\alpha$ -amylase
- D. Lingual lipase
- E. Lysozyme

**3. A neurologist tests a patient with an isolated lesion of the nerve that carries parasympathetic drive to the parotid gland. Which nerve is affected?**

EASY

- A. Trigeminal nerve (CN V)
- B. Hypoglossal nerve (CN XII)
- C. Facial nerve (CN VII)
- D. Glossopharyngeal nerve (CN IX)
- E. Vagus nerve (CN X)

**4. A respiratory therapist notices that a patient's breathing stops briefly every time he swallows. This is best described as:**

EASY

- A. Deglutitive inhibition
- B. Deglutition apnea
- C. A cephalic-phase response
- D. Secondary peristalsis
- E. Aspiration of the bolus

**5. A 45-year-old man has had his parotid gland removed. During a meal, when saliva flow is strongly stimulated, what change is most likely compared with a person with intact glands?**

MEDIUM

- A. No change, because the submandibular gland supplies all of the stimulated saliva
- B. Entirely mucous saliva, because the sublingual gland takes over serous secretion
- C. No saliva at rest, because the parotid is the main source of unstimulated saliva
- D. A larger rise in amylase-rich saliva, because the submandibular gland becomes serous
- E. A smaller rise in flow and less amylase-rich saliva, because parotid output rises strongly with stimulation

**6. Before a viva, a student's mouth feels dry even though some saliva is still being secreted. Which mechanism best explains this?**

MEDIUM

- A. Strong parasympathetic drive via CN VII and CN IX, giving copious watery saliva
- B. Blockade of muscarinic receptors by an antimuscarinic drug taken earlier that day
- C. A cephalic-phase response to the sight and smell of food
- D. Sympathetic outflow via the superior cervical ganglion, giving smaller-volume, protein-rich saliva
- E. Parasympathetic vasodilation that increases glandular blood flow and secretion

**7. A cook smells frying onions and her mouth waters before she has tasted anything. Which statement best explains this?**

MEDIUM

- A. It is a cephalic-phase response: conditioned cues can raise saliva before food enters the mouth
- B. It occurs because the salivatory nuclei are inhibited when the food is smelled
- C. It reflects reduced fluid availability in the glands after a period of dehydration
- D. Salivation can only begin once food touches the oral mucosa and stimulates taste receptors
- E. It is a purely sympathetic response producing thick, scanty saliva in anticipation

**8. A researcher compares saliva collected at a slow resting flow with saliva collected during vigorous stimulation. Why is the stimulated sample closer to the isotonic primary secretion?**

MEDIUM

- A. At higher flow the ducts modify the fluid more extensively
- B. At higher flow there is less ductal modification of the acinar fluid
- C. Stimulation removes mucins, making the fluid isotonic
- D. At higher flow the acini secrete a hypotonic fluid
- E. Stimulation switches off the acini and ducts secrete the fluid instead

**9. A patient with an oropharyngeal problem reports that fluids come out of his nose when he swallows. Which normal step of the pharyngeal phase is most likely failing?**

MEDIUM

- A. Primary peristalsis of the esophageal body
- B. Relaxation of the upper esophageal sphincter
- C. Relaxation of the lower esophageal sphincter ahead of the bolus
- D. Elevation of the soft palate to close the nasopharynx
- E. Adduction of the vocal folds to close the glottis

**10. A neurologist maps the swallowing reflex circuit. Which nerves carry afferent information about bolus location and touch from the oral cavity and pharynx to the brainstem?**

MEDIUM

- A. CN IX, X and XII
- B. CN VII, XI and XII
- C. CN V, VII, IX, X and XII
- D. CN III, IV and VI
- E. CN V, IX and X

**11. After a bolus has been swallowed, a fragment of food remains lodged in the mid-esophagus. Which mechanism moves it on?**

MEDIUM

- A. Secondary peristalsis, triggered by esophageal distension
- B. Deglutition apnea
- C. Contraction of the upper esophageal sphincter
- D. Gravity alone
- E. A second, voluntary tongue push that re-triggers primary peristalsis

**12. In an animal experiment, the vagal and enteric inhibitory pathways that use NO and VIP are blocked. What would happen when the animal swallows?**

MEDIUM

- A. Deglutition apnea would be abolished
- B. The upper esophageal sphincter would fail to open
- C. Salivary flow would stop
- D. The LES would fail to relax ahead of the bolus
- E. Primary peristalsis would be converted into secondary peristalsis

**13. A 62-year-old man says food sticks behind his sternum after swallowing, but only when he eats solid food; liquids pass easily. Which problem is most likely?**

MEDIUM

- A. An esophageal motility disorder
- B. Failure of the brainstem swallowing network
- C. Mechanical obstruction of the esophageal lumen
- D. Failure of airway protection in the pharyngeal phase
- E. Reduced salivary flow from an antimuscarinic drug

**14. A dentist wants to list the salivary components that contribute to oral defence. Which set contains only defence components?**

MEDIUM

- A. Lingual lipase,  $\alpha$ -amylase, mucins and growth factors
- B.  $\alpha$ -amylase, mucins, urea and uric acid
- C.  $\text{Na}^+$ ,  $\text{Cl}^-$ , urea and uric acid
- D. Bicarbonate,  $\text{Ca}^{2+}$ , phosphate and  $\alpha$ -amylase
- E. IgA, lysozyme, lactoferrin and peroxidases

**15. A lesion selectively interrupts the parasympathetic supply of CN IX on one side; CN VII is intact. What is the most likely finding when chewing strongly stimulates flow?**

HARD

- A. Reduced secretion from the submandibular and sublingual glands, with a normal parotid response
- B. Reduced watery, amylase-rich secretion from the parotid, with the submandibular and sublingual glands still responding via CN VII
- C. Increased parotid secretion, because the unopposed sympathetic supply drives copious saliva
- D. Loss of all salivation on that side, because every salivary gland depends on CN IX for its drive
- E. No change, because the sympathetic supply replaces parasympathetic drive with equally copious watery saliva

**16. After neck surgery a patient has lost his sympathetic supply to the salivary glands (superior cervical ganglion destroyed) but parasympathetic pathways are intact. What is expected during a meal?**

HARD

- A. No saliva at all, because both divisions are needed to start secretion
- B. Parasympathetic vasodilation of the glands is abolished
- C. Loss of the copious watery secretion, since parasympathetic drive depends on the ganglion
- D. Saliva becomes more protein-rich than normal
- E. Copious watery saliva still occurs, but the smaller-volume, protein-rich sympathetic component is lost

**17. A disease selectively destroys the ductal cells of the salivary glands while the acinar cells remain intact. Compared with a healthy person at the same flow rate, the saliva would most likely be:**

HARD

- A. Devoid of water, because ducts supply the fluid
- B. Closer to the isotonic primary secretion, because ductal modification is lost
- C. More hypotonic than normal, because the acini secrete a hypotonic fluid
- D. More hypotonic, because ductal modification is exaggerated
- E. Unchanged, because the acini alone determine final composition

**18. A 73-year-old man after a stroke shows a normally tilting epiglottis on imaging but repeatedly aspirates thin liquid. Which explanation best fits the lecture?**

HARD

- A. The epiglottis alone does not seal the airway; other protective layers (glottic closure, laryngeal elevation, timing) have failed
- B. The soft palate has failed to elevate, so the bolus is entering the nasopharynx
- C. Aspiration is impossible if the epiglottis tilts, so failed esophageal peristalsis must be the cause
- D. Salivary flow is too low to lubricate the bolus, so it cannot be moved safely
- E. The LES has failed to relax ahead of the bolus, causing reflux into the airway

**19. A 50-year-old has had progressive difficulty with both solids and liquids since the start, a sense of food stuck behind the sternum, and no coughing when he begins to swallow. Which explanation fits best?**

HARD

- A. Xerostomia from reduced watery saliva
- B. A mechanical stricture, which would affect solids before liquids
- C. Loss of inhibitory myenteric neurons, so the LES fails to relax and peristalsis is absent or abnormal
- D. Oropharyngeal dysphagia from failed airway protection, which would cause immediate choking
- E. Failure of the soft palate to close the nasopharynx

**20. After a brainstem stroke a patient can chew, mix the food with saliva and push it to the back of his mouth, but the swallow then fails to trigger properly and he coughs. Which best describes the deficit?**

HARD

- A. The voluntary oral phases are preserved, but the automatic pharyngeal sequence is impaired
- B. Parasympathetic drive to the salivary glands has been lost, so food cannot be moistened
- C. The deficit is confined to the LES, which fails to relax ahead of the bolus
- D. Primary peristalsis is intact, so the pharyngeal phase must also be intact
- E. The oral preparatory phase has failed, because chewing depends on the brainstem swallowing network

## ✓ Answer key

| Q  | Answer   | Level  | Why (lecture slide)                                                                                                                                                                                                        |
|----|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>C</b> | EASY   | The submandibular gland (mixed serous + mucous) is the major source of unstimulated saliva. [Slide 3]                                                                                                                      |
| 2  | <b>C</b> | EASY   | Salivary $\alpha$ -amylase begins starch digestion; lingual lipase contributes to fat digestion. [Slide 5]                                                                                                                 |
| 3  | <b>D</b> | EASY   | CN IX $\rightarrow$ parotid; CN VII $\rightarrow$ submandibular + sublingual. [Slide 6]                                                                                                                                    |
| 4  | <b>B</b> | EASY   | Breathing briefly pauses during the pharyngeal phase (deglutition apnea) as part of airway protection. [Slides 10, 13]                                                                                                     |
| 5  | <b>E</b> | MEDIUM | Parotid = mostly serous, amylase-rich, contribution rises strongly during stimulation; the submandibular gland is the resting source. [Slide 3]                                                                            |
| 6  | <b>D</b> | MEDIUM | Fear/anxiety gives a dry-mouth sensation; sympathetic states give smaller-volume, protein-rich saliva and can feel "dry". [Slides 6, 7]                                                                                    |
| 7  | <b>A</b> | MEDIUM | Smell + sight (conditioned responses) increase salivation; the cephalic phase means saliva can rise before food enters the mouth. [Slide 7]                                                                                |
| 8  | <b>B</b> | MEDIUM | Acini make a primary isotonic fluid; ducts modify it to mostly hypotonic saliva; higher flow = less ductal modification. [Slide 20]                                                                                        |
| 9  | <b>D</b> | MEDIUM | The soft palate elevates $\rightarrow$ nasopharynx closes; nasal regurgitation is a clue to oropharyngeal dysphagia. [Slides 13, 18]                                                                                       |
| 10 | <b>E</b> | MEDIUM | Sensory input = V, IX, X; motor output = V, VII, IX, X, XII. [Slide 15]                                                                                                                                                    |
| 11 | <b>A</b> | MEDIUM | Secondary peristalsis is triggered by esophageal distension if material remains; gravity helps but is not required. [Slide 16]                                                                                             |
| 12 | <b>D</b> | MEDIUM | The LES relaxes before the bolus arrives via deglutitive inhibition (vagal + enteric inhibitory pathways, NO/VIP). [Slides 16, 19]                                                                                         |
| 13 | <b>C</b> | MEDIUM | Esophageal dysphagia: solids only $\rightarrow$ mechanical obstruction; solids + liquids $\rightarrow$ motility disorder. [Slide 18]                                                                                       |
| 14 | <b>E</b> | MEDIUM | Protection: mechanical cleansing + IgA, lysozyme, lactoferrin, peroxidases (+ antimicrobial peptides). [Slides 4, 5]                                                                                                       |
| 15 | <b>B</b> | HARD   | CN IX drives the parotid (serous, amylase-rich, rises with stimulation); CN VII drives submandibular + sublingual. [Slides 3, 6]                                                                                           |
| 16 | <b>E</b> | HARD   | Parasympathetic activity $\rightarrow$ copious watery saliva + vasodilation; sympathetic (via superior cervical ganglion) $\rightarrow$ smaller-volume, protein-rich saliva. [Slide 6]                                     |
| 17 | <b>B</b> | HARD   | Acini make a primary isotonic fluid; ducts modify it into mostly hypotonic saliva. [Slide 20]                                                                                                                              |
| 18 | <b>A</b> | HARD   | Airway protection is layered; epiglottis alone does NOT seal the airway. Post-stroke coughing after water = pharyngeal dysphagia. [Slides 14, 17]                                                                          |
| 19 | <b>C</b> | HARD   | Solids + liquids $\rightarrow$ motility disorder; achalasia = loss of inhibitory myenteric neurons (NO/VIP) $\rightarrow$ LES fails to relax; esophageal dysphagia has no immediate choking at initiation. [Slides 18, 19] |
| 20 | <b>A</b> | HARD   | Oral phases are voluntary; the pharyngeal sequence is automatic, coordinated by the brainstem network (NTS, nucleus ambiguus, central pattern generator). [Slides 11, 15]                                                  |